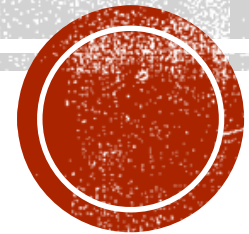


RECONFIGURABLE COMPUTING

CS G553: Reconfigurable computing

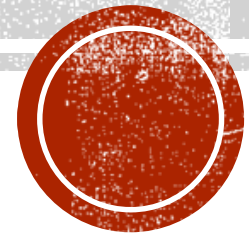
Dr. Meetha V Shenoy

Lect-13

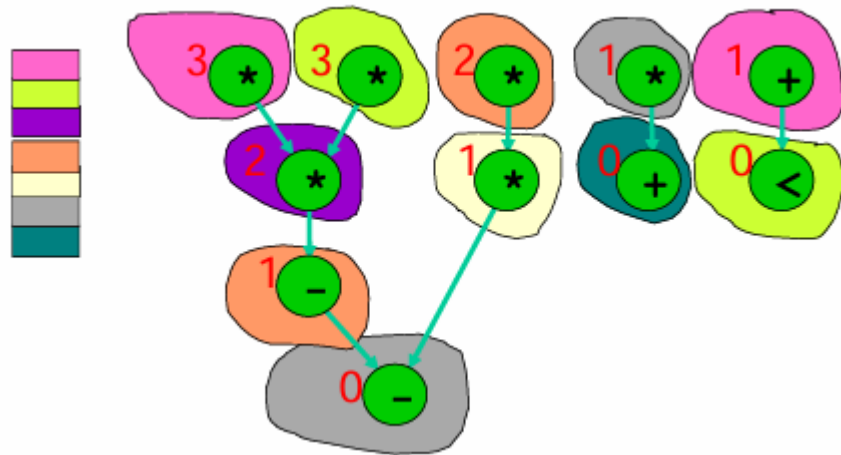


ARCHITECTURAL SYNTHESIS (HIGH LEVEL SYNTHESIS)

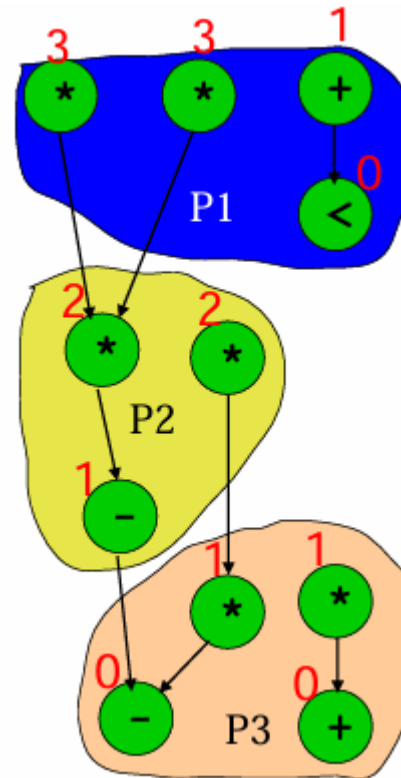
TEMPORAL PARTITIONING & SCHEDULING



TEMPORAL PARTITIONING



- Criterion: number of successors
- size(FPGA) = 250,
- size (mult) = 100,
- size(add) = size(sub) = 20,
- size(comp) = 10.



TEMPORAL PARTITIONING IN RCS

In RCS,

- Resource types are not important.
- The amount of basic resources (LUT) is important.
- Operators do not compete for resources. - They compete for area.
- The starting time and the end time of the complete partitions are usually considered



TEMPORAL PARTITIONING IN RCS-LIST SCHEDULING

- **Assignment criterion:** there should be enough places left on the device to accommodate the new component.

Algorithm: List-scheduling algorithm for reconfigurable devices

```
sort the nodes of  $v$  according to their priorities
 $P_0 := \emptyset$ 
while  $V \neq \emptyset$  do
    select a vertex  $v \in V$  with highest priority and whose predecessors
are all placed
    if (a partition  $P_i$  exists with  $s(P_i) + s(v) \leq s(H)$ ) then
         $P_i = P_i \cup \{v\}$ 
    else
        create a new partition  $P_{i+1}$  and set  $P_{i+1} = \{v\}$ 
    end if
end while
```



TEMPORAL PARTITIONING

- o Total computation time of DFG:

$$t_{DFG} = n \times C_H + \sum_{1,...,n}(t_{P_i})$$

- o C_H : Reconfiguration time of device H
- o t_{P_i} : Computation time of partition P_i .
- o n : Number of partitions

Optimization:

- o If C_H too large, then the optimization will tend to minimize the number of partitions
- o If $C_H \ll t_p$, then algorithm will tend to avoid long paths in partitions.



TEMPORAL PARTITIONING

Advantages of LS-based temporal partitioning:

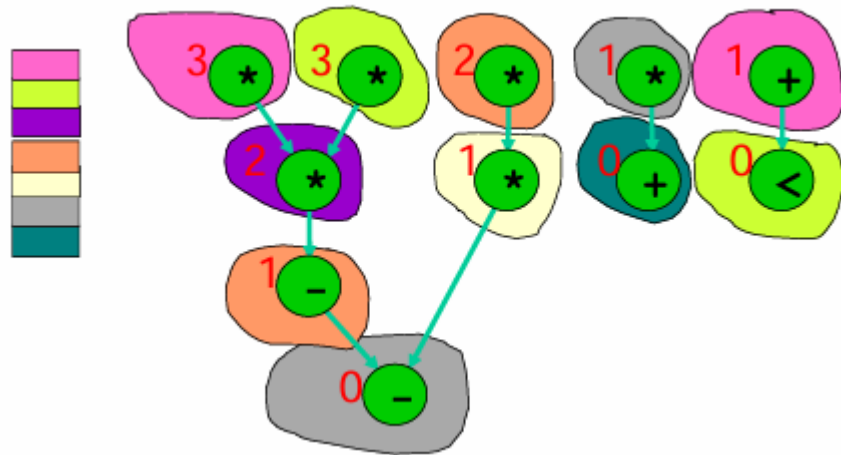
- o Fast (linear time algorithm)

- **Disadvantage:**

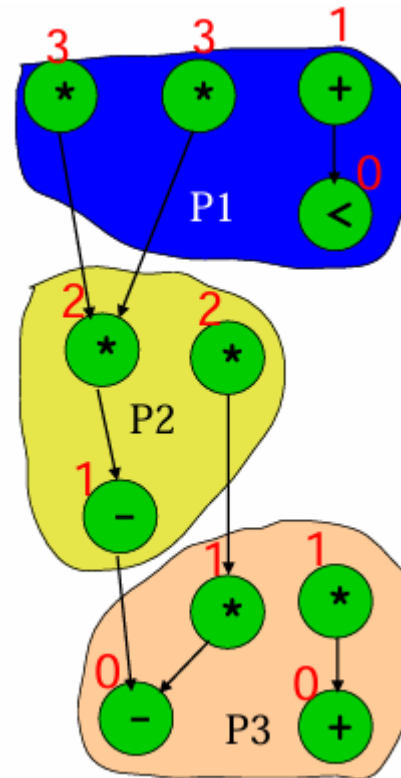
- ➤ Levelization: Modules are assigned to partitions based more on their **level number** rather than their interconnectivity with another component.



TEMPORAL PARTITIONING



- Criterion: number of successors
- size(FPGA) = 250,
- size (mult) = 100,
- size(add) = size(sub) = 20,
- size(comp) = 10.

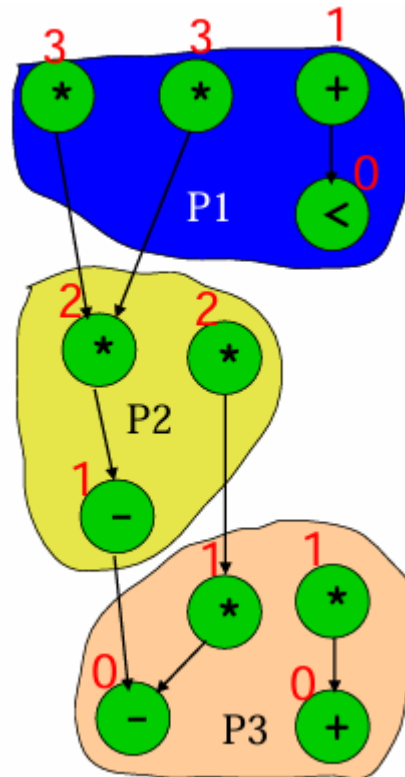


- Connectivity:
 - $c(P1) = 1/6$,
 - $c(P2) = 1/3$,
 - $c(P3) = 2/6$.
- Quality: 0.28



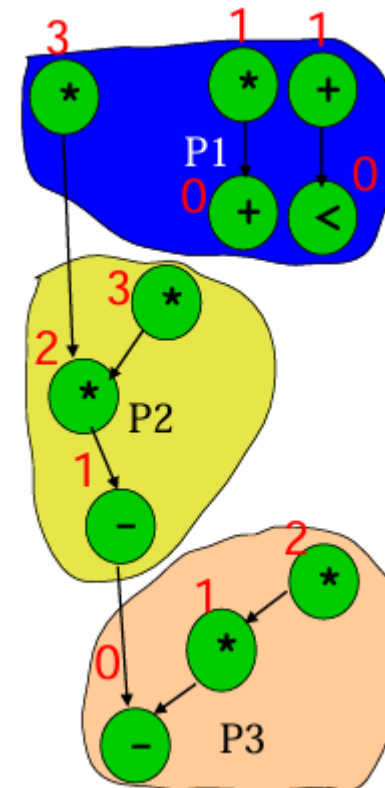
TEMPORAL PARTITIONING- IMPROVEMENT

- Criterion: number of successors
- size(FPGA) = 250,
- size (mult) = 100,
- size(add) = size(sub) = 20,
- size(comp) = 10.



- Connectivity:
 - $c(P1) = 1/6$,
 - $c(P2) = 1/3$,
 - $c(P3) = 2/6$.

• Quality: 0.28



- Connectivity:
 - $c(P1) = 2/10$,
 - $c(P2) = 2/3$,
 - $c(P3) = 2/3$.

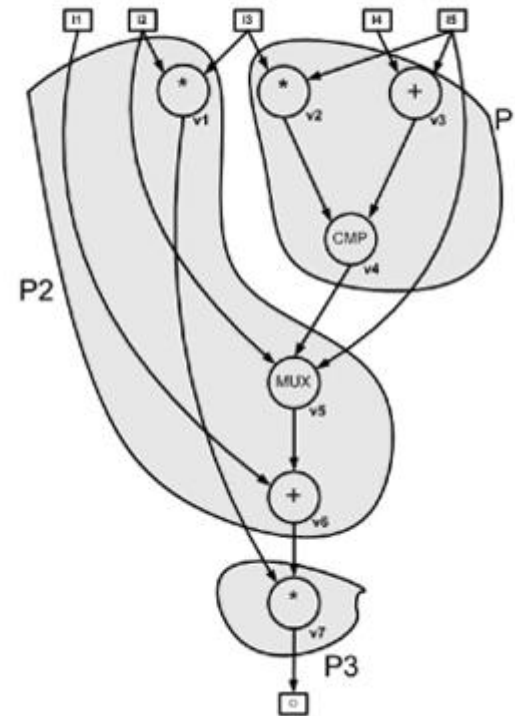
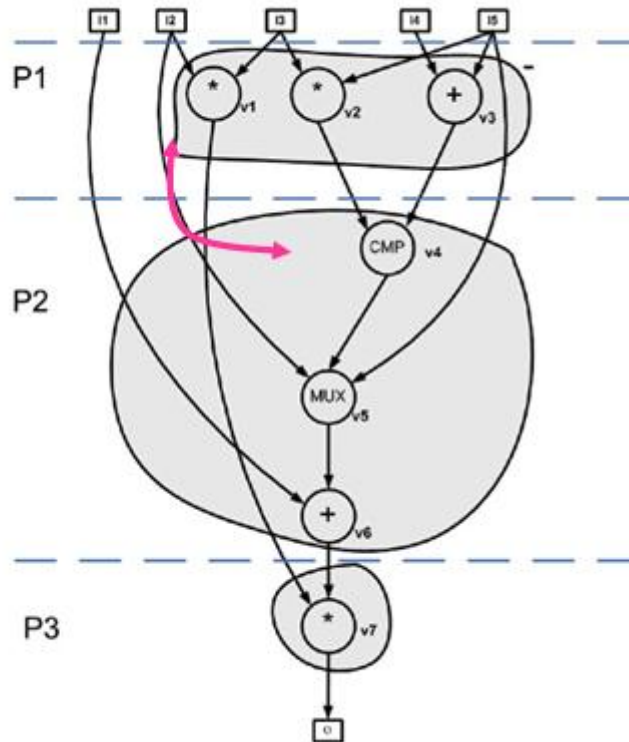
• Quality: 0.51

▪ Quality is better



TEMPORAL PARTITIONING- IMPROVEMENT

□ Pair wise interchange



TEMPORAL PARTITIONING

Advantages of LS-based temporal partitioning:

- o Fast (linear time algorithm)

- **Disadvantage:**

- ➤ Levelization: Modules are assigned to partitions based more on their level number rather than their interconnectivity with another component.

Improvements

→ Interconnectivity (data exchange) must be optimized.

→ Local optimization possible • **e.g. configuration switching**

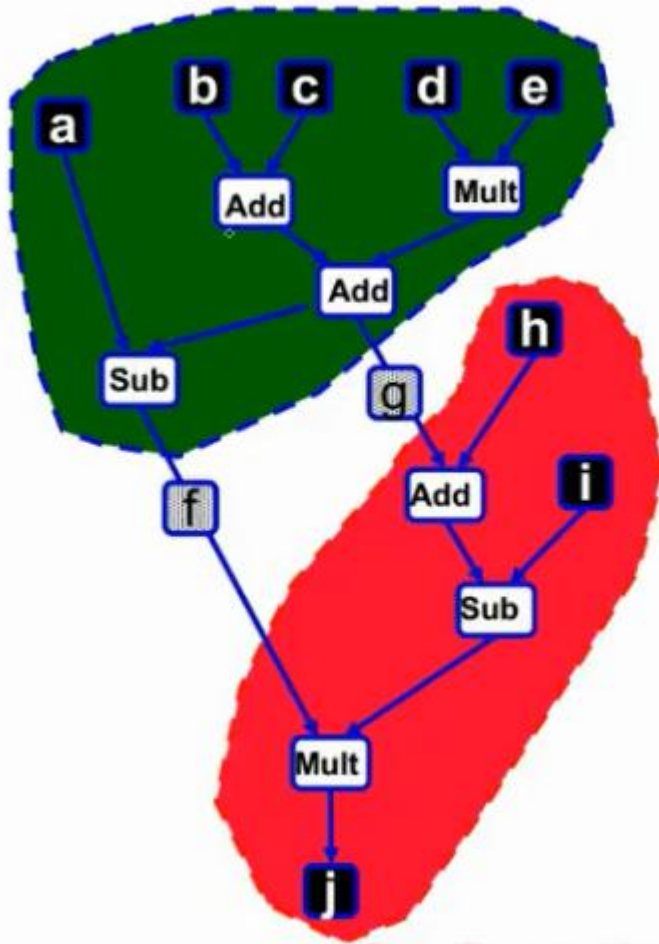


CONFIGURATION SWITCHING

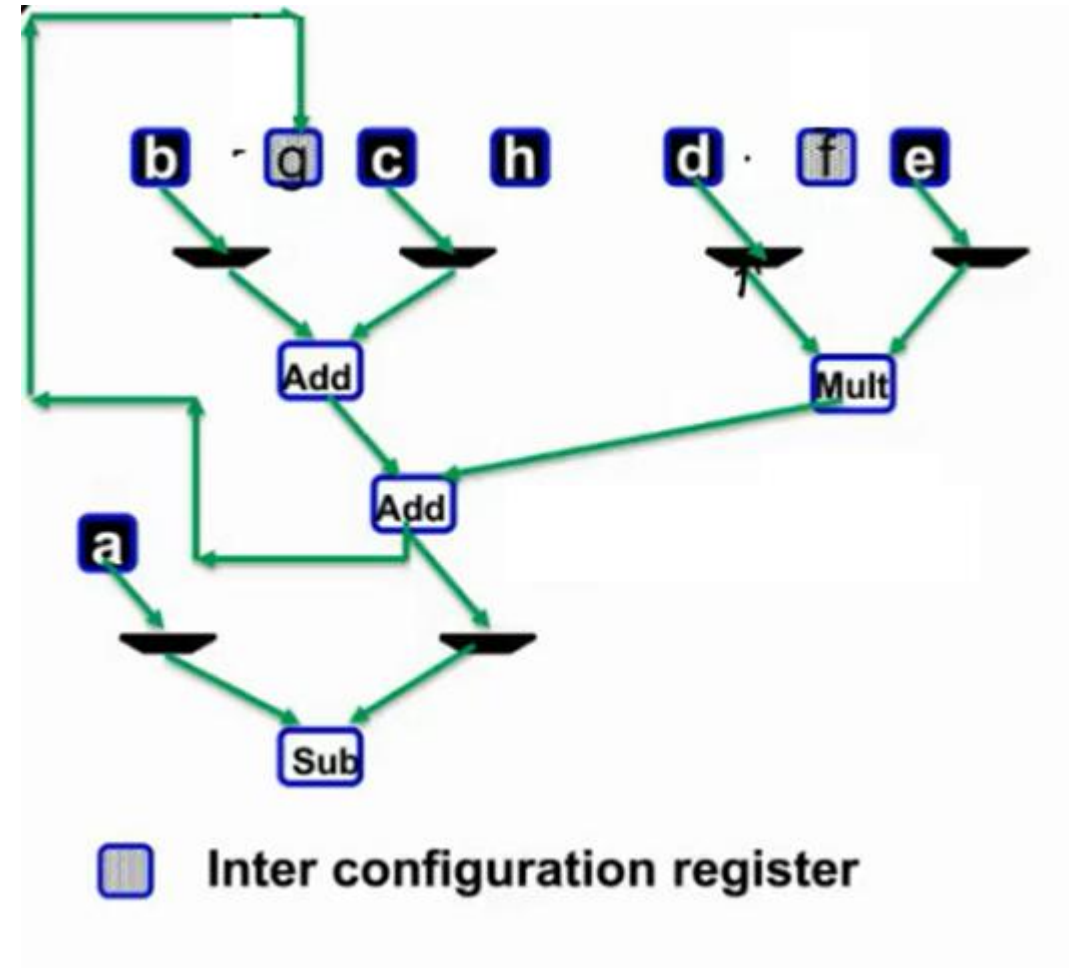
- If two consecutive partitions P1 and P2 have a common set of operators, then can use **partial reconfiguration** provided signal multiplexing to switch from one partition to another.



Configuration 1



Configuration 2



 Inter configuration register



```

graph TD
    a[a] --> Sub1[Sub]
    b[b] --> Add1[Add]
    c[c] --> Add1
    d[d] --> Mult1[Mult]
    e[e] --> Mult1
    Add1 --> Add2[Add]
    Mult1 --> Add2
    Add2 --> Sub1
    Sub1 --> f[f]
    f --> Mult2[Mult]
    h[h] --> Add3[Add]
    i[i] --> Sub2[Sub]
    Add3 --> Sub2
    Sub2 --> Mult2
    Mult2 --> j[j]
    
```